

Question Bank

MSc. C.A. SEM-3 : 904-01 Application Development using UI

Short Questions

1. Differentiate between Single-page Applications (SPA) and Multi-page Applications (MPA).
2. What is Virtual DOM? How does it work in React?
3. Write any two differences between ES5 and ES6.
4. What is the difference between Props and State in React?
5. List any two lifecycle methods of a React Class Component.
6. What is **react-router**? Mention any two of its features.
7. What is the purpose of a **404 Page** in React routing?
8. Differentiate between **Link** and **NavLink** in React Router.
9. What is the use of **URL Parameters** in React Router? Give an example.
10. What is **json-server** and why is it used in React development?
11. What is **Redux** and why is it needed in React applications?
12. Explain the role of a **Reducer** in Redux.
13. List the three core principles of Redux.
14. What is the purpose of **Redux-Saga** middleware?
15. What are **Generators** in Redux-Saga and why are they used?
16. List any two pros of **GraphQL** over REST API.
17. What is the **type system** in GraphQL?
18. Name any two **GraphQL data types**.
19. What is the use of the **GraphiQL tool**?
20. What is the role of **Apollo Client** in GraphQL applications?

Long Questions

1. Explain the differences between **Single-Page Applications (SPAs)** and **Multi-Page Applications (MPAs)**, focusing on their architecture and user experience.
2. What are **Props** and **State** in React? Show examples using both **Class-based** and **Function-based** components.
3. Explain how **routing** is implemented in a React application using **react-router**.
4. How can you build a **mock REST API** using **json-server**? Write steps to consume an API in a React application using the **Fetch** method.

5. Describe the **Redux Architecture** with all its components: **Action**, **Reducer**, **Store**.
6. Define **Redux-Saga**. What are **Generators** in Redux-Saga?
7. Explain the process of building a **full-stack application** using **GraphQL**.
8. Explain how to set up a **testing environment using Jest** for a React application.
9. What is **Virtual DOM** in React? Explain its working mechanism and how it improves performance compared to the Real DOM.
10. Differentiate between **ES5** and **ES6** with at least five key features and relevant code examples.
11. Explain the **lifecycle** of a React **Class Component** with the help of a diagram and examples of lifecycle methods.
12. What is the difference between **Link** and **NavLink** in React Router? Demonstrate with an example how **NavLink** is used to apply active styles.
13. Describe **URL Parameters** and **Nested Routes** in React Router with an example.
14. Explain the **need for asynchronous operations** in Redux. Describe the **async workflow**, **action creators**, and how **reducers** handle async actions.
15. What are the **core principles of Redux**? Explain each principle with an example.
16. Explain the **features of GraphQL** compared to REST API, and describe how frontend-backend communication works using GraphQL.
17. Describe the **Apollo Client** setup process in a React application and explain how it communicates with a GraphQL server.
18. Explain the use of **GraphiQL tool** and describe how **schemas** and **types** are defined in a GraphQL API.

MCQ Questions

1. Which of the following best describes a Single-page Application (SPA)?

- A) An application that reloads the entire page on every user interaction
- B) An application that loads all resources at once and updates the view dynamically without full page reloads
- C) An application that has only one HTML file without JavaScript
- D) An application that is built using only server-side rendering

Answer: B

2. Which architecture pattern separates the application into Model, View, and Controller?

- A) MVVM
- B) MVP
- C) MVC
- D) MVT

Answer: C

3. In React, which of the following is TRUE about the Virtual DOM?

- A) It directly manipulates the browser DOM
- B) It is a lightweight copy of the actual DOM stored in memory
- C) It replaces HTML completely
- D) It is a database for storing components

Answer: B

4. Which ES6 feature allows declaring variables with block scope?

- A) var
- B) let
- C) const
- D) Both B and C

Answer: D

5. In React, which lifecycle method is called only once immediately after the component is inserted into the DOM?

- A) componentDidMount()
- B) componentWillUnmount()
- C) render()

D) shouldComponentUpdate()

Answer: A

6. The full form of ECMA is:

- A) European Communication Management Association
- B) European Computer Manufacturers Association
- C) Electronic Code Management Association
- D) European Coding Manufacturers Association

Answer: B

7. Which library is commonly used for routing in React applications?

- A) react-navigation
- B) react-router
- C) router-js
- D) react-path

Answer: B

8. In React Router, which component is used for applying styles to active links?

- A) Link
- B) NavLink
- C) StyledLink
- D) ActiveLink

Answer: B

9. What is the main use of URL parameters in React Router?

- A) To load CSS files dynamically
- B) To pass dynamic values through the URL
- C) To store JavaScript variables
- D) To cache pages for offline use

Answer: B

10. Which command is used to install json-server?

- A) npm install json-server
- B) npm create json-server
- C) npx create json-server

D) npm add json-server-cli

Answer: A

11. Which JavaScript method is commonly used to fetch data from an API in React?

- A) request()
- B) load()
- C) fetch()
- D) getData()

Answer: C

12. In Redux, the central place where the application state is stored is called:

- A) Reducer
- B) Store
- C) Action
- D) Middleware

Answer: B

13. Which of the following is NOT a principle of Redux?

- A) Single source of truth
- B) State is read-only
- C) Changes are made with pure functions
- D) Multiple stores for different components

Answer: D

14. In Redux, which function returns an object describing what happened in the application?

- A) Reducer
- B) Store
- C) Action Creator
- D) Dispatcher

Answer: C

15. Which method in Redux-Saga listens for specific dispatched actions and triggers a saga function?

- A) takeEvery()
- B) put()

- C) call()
- D) delay()

Answer: A

16. Which browser tool is commonly used for debugging Redux state changes?

- A) React DevTools
- B) Redux DevTools
- C) Saga Debugger
- D) State Inspector

Answer: B

17. Which of the following is an advantage of GraphQL over REST?

- A) It always returns fixed data structures
- B) It allows clients to request exactly the data they need
- C) It uses only XML format for responses
- D) It eliminates the need for a backend server

Answer: B

18. Which data type in GraphQL represents a sequence of characters?

- A) String
- B) Int
- C) Boolean
- D) ID

Answer: A

19. Which tool provides an interactive environment to test GraphQL queries?

- A) Postman
- B) GraphiQL
- C) Apollo Studio
- D) GraphQL Playground

Answer: B

20. In GraphQL, which framework is commonly used to build a server-side API in Node.js?

- A) Express
- B) Django

- C) Laravel
- D) Flask

Answer: A

21. Which of the following is required to set up Apollo Client in a React application?

- A) npm install apollo-server
- B) npm install @apollo/client graphql
- C) npm install graphql-server
- D) npm install apollo-react

Answer: B